

Student Management Hub

- **Problem Statement:** How can a student manage his/her grades and schedule with a user-friendly digital tool?
- **Project Objectives:**
 - The objective of our project is to create a program that can help students keep track of their grades and due dates. The program will be able to:
 - Record a student's quiz and test scores
 - Use inputted requirement scores to compute a student's grade for a subject
 - Compute a student's general weighted average using the calculated subject grades
 - Help students organise their schedules by tracking project due dates and the completion status of such projects
- **Planned Features:**
 - Account system:
 - Allows multiple students to use the program on a single computer without risking scheduling conflicts
 - Each account will have its own file where the corresponding student's data will be written
 - Student Grade Management:
 - Allows students to record their scores, view their grades, and calculate their quarterly averages
 - Schedule Tracker
 - The program will be able to track due dates and upcoming events, as well as notify the user of any imminent deadlines
- **Planned Inputs and Outputs:**
 - **Inputs:**
 - Login interface: student account information
 - Menu interfaces: the user's chosen option
 - Student Grade Management:
 - Student scores for the quarterly average calculation
 - Schedule Tracker:
 - Task information
 - Due dates
 - **Outputs:**
 - Menu prompts
 - Student Grade Management:
 - Calculated subject grades
 - Calculated GWA
 - Schedule Tracker:
 - To-do list (remaining requirements)
 - Calendar with deadlines indicated

- Time remaining until each due date
- Subjects for the current and following days

- **Logic Plan:**

```
DEFINE gradeManagement():
```

```
    DECLARE INTEGER choice, month
```

```
    DECLARE FLOAT score, maximum
```

```
    DECLARE STRING subject, name, type
```

```
    WHILE True:
```

```
        OUTPUT "1. Record grades"
```

```
        OUTPUT "2. Display grade summary"
```

```
        OUTPUT "3. Exit"
```

```
        INPUT choice
```

```
        CASE choice OF:
```

```
            1:
```

```
                WHILE True:
```

```
                    OUTPUT "How will you enter this activity's information?"
```

```
                    OUTPUT "1. Manual entry"
```

```
                    OUTPUT "2. Select an activity recorded from the schedule
```

```
tracker"
```

```
                    INPUT choice
```

```
                    CASE choice OF:
```

```
                        1:
```

```
                            OUTPUT "Is this an FA or an AA?"
```

```
                            INPUT type
```

```
                            OUTPUT "Enter the subject the activity falls
```

```
under... "
```

```
                            INPUT subject
```

```
                            OUTPUT "Enter a name for the activity... "
```

```
                            INPUT name
```

```
                            OUTPUT "What's this activity's maximum
```

```
possible score?"
```

```
                            INPUT maximum
```

```
                            OUTPUT "What's your score on this
```

```
activity? "
```

```
                            INPUT score
```

```
                            [The data is written to the student's file.]
```

```
                        2:
```

```
                            OUTPUT "Enter a month to display records
```

```
for it: (numerical form, 1 - 12)"
```

```
                            INPUT month
```

[Using a while loop, the activities for that month will be displayed one by one, each with a number.]
OUTPUT "Enter the number corresponding to the activity... "

INPUT choice

[If no such activity is bound to the provided number, a message will be displayed and a continue statement will be run. Otherwise:]

OUTPUT "What's this activity's maximum possible score?"

INPUT maximum

OUTPUT "What's your score on this activity? "

INPUT score

[The data is then written to the student's file.]

OUTPUT "Would you like to record another activity? (y/n)"

INPUT choice

IF choice == "n" THEN

BREAK

END IF

2:

[Activity scores are read from the student's file.]

[Each subject's grades for the quarter are calculated. (This will likely make use of a for loop iterating through a list of the student's subjects and a subloop with accumulator variables.)]

[The PSHS grade formula is used to compute the quarterly average with the calculated grades.]

[The raw average is then transmuted using the previous quarter's average. (In the first quarter, it is left as it is.)]

[Finally, the subject grades and GWA are printed to the console.]

3:

BREAK

default:

OUTPUT "Please enter a valid input."

DEFINE scheduler():

DECLARE INTEGER choice

DECLARE STRING name, date, type

WHILE True:

[If there is an activity that is due in a week or less, a notice will be displayed with information about the activity and how much time remains before it is due.]

OUTPUT "1. Record due dates"

OUTPUT "2. Display due date summary"

OUTPUT "3. Exit"

INPUT choice

CASE choice OF:

1:

WHILE True:

OUTPUT "How will you enter this activity's information?"

OUTPUT "1. Manual entry"

OUTPUT "2. Select an activity recorded from the grade management system"

INPUT choice

CASE choice OF:

1:

OUTPUT "Is this an FA or an AA?"

INPUT type

OUTPUT "Enter the subject the activity falls under... "

INPUT subject

OUTPUT "Enter a name for the activity... "

INPUT name

OUTPUT "When is this due? (mm/dd/yyyy) "

INPUT date

[The data is written to the student's file.]

2:

OUTPUT "Enter a month to display records for it: (numerical form, 1 - 12)"

INPUT month

[Using a while loop, the activities for that month will be displayed one by one, each with a number.]

OUTPUT "Enter the number corresponding to the activity... "

INPUT choice

[If no such activity is bound to the provided number, a message will be displayed and a continue statement will be run. Otherwise:]

OUTPUT "When is this due?
(mm/dd/yyyy) "
INPUT date
[The data is then written to the student's file.]

OUTPUT "Would you like to record another activity? (y/n)"
INPUT choice
IF choice == "n" THEN
 BREAK
END IF

2:
 [Through nested for loops, the due dates for each activity will be outputted, categorized by month.]
 [Another for loop will be used to print the days remaining until each of the due dates.]

3:
 BREAK

default:
 OUTPUT "Please enter a valid input."

```
DEFINE main():  
    DECLARE INTEGER choice  
    WHILE True:  
        OUTPUT "Menu:"  
        OUTPUT "1. Grade Management System"  
        OUTPUT "2. Schedule Tracker"  
        OUTPUT "3. Exit"  
  
        CASE choice OF:  
            1: CALL gradeManagement()  
            2: CALL scheduler()  
            3: BREAK  
            default: OUTPUT "Please enter a valid input."  
  
CALL main()
```